

Lecture 7

Chapter 23

Gauss' Law

□ Applying Gauss' Law: Cylindrical Symmetry

Let, an infinitely long cylindrical plastic rod with uniform charge density, λ . Consider a smallest region with a tiny area, $d\mathbf{A}$ on the upper surface of the cylindrical Gaussian surface having radius r , and height, h (area, $2\pi r h$) which enclosed the charges, q_{enc}

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E dA = \frac{q_{enc}}{\epsilon_0}$$

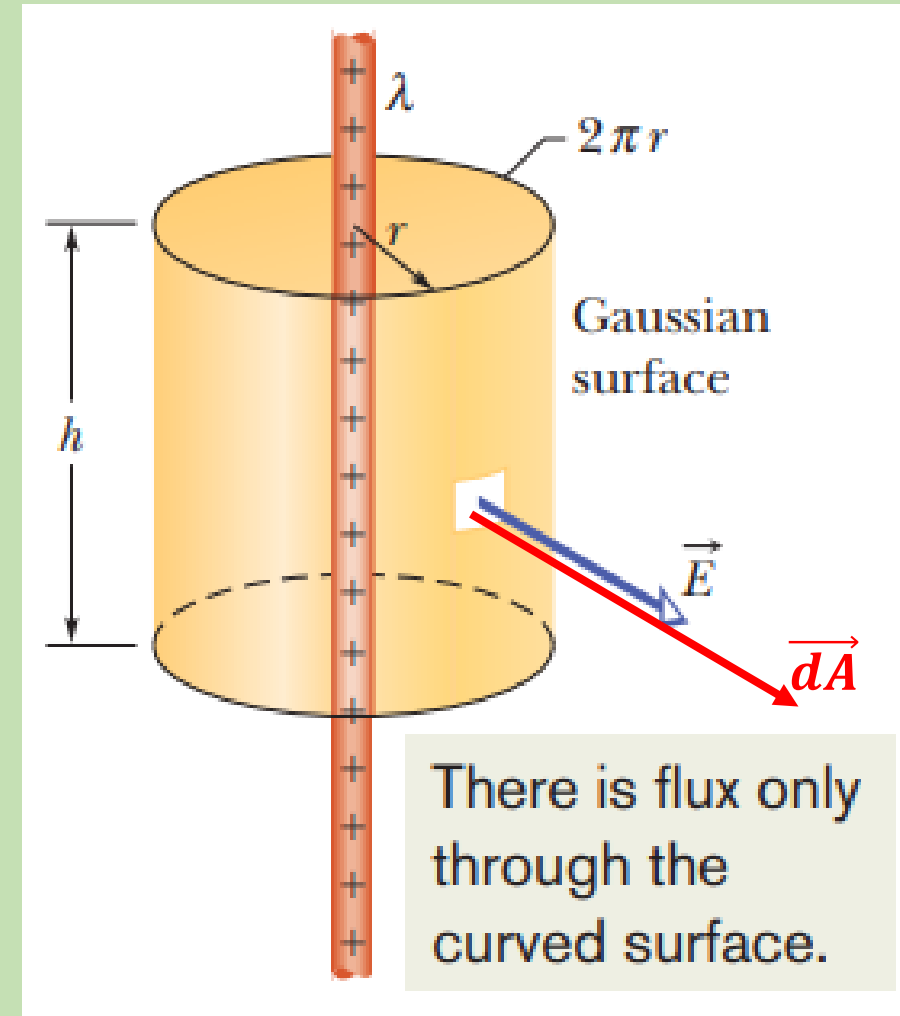
$$\Rightarrow E \oint dA = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E(2\pi r h) = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow EA = \frac{q_{enc}}{2\pi r h \epsilon_0}$$

$$\Rightarrow E = \frac{\frac{q_{enc}}{h}}{2\pi \epsilon_0 r}$$

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$



□ Applying Gauss' Law: Spherical Symmetry

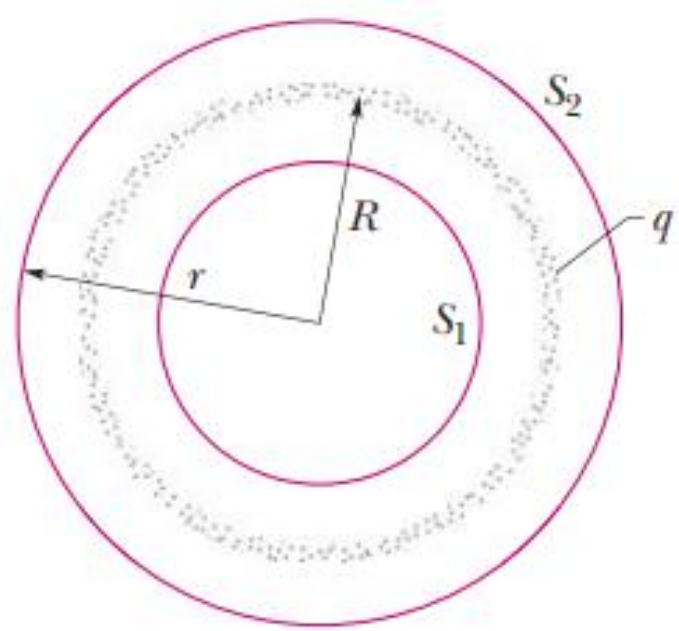
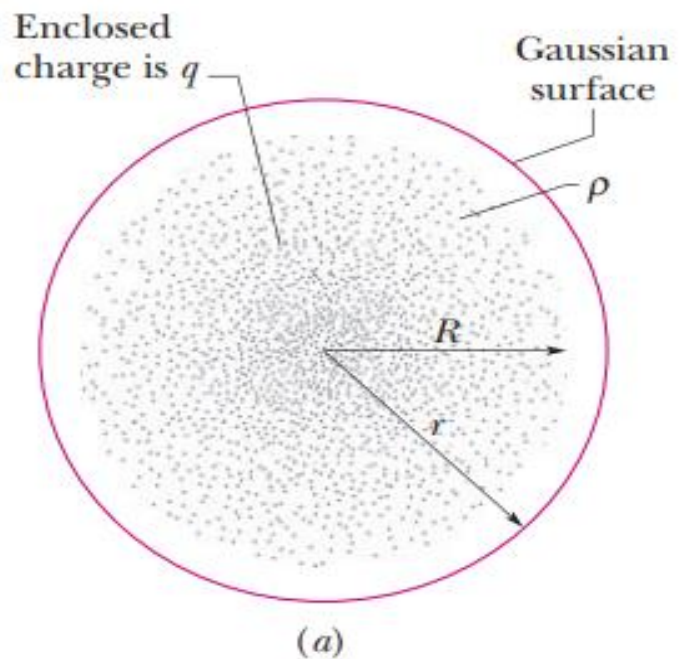


Figure shows a charged spherical shell of total charge q and radius R and two concentric spherical Gaussian surfaces, S_1 and S_2 .

If we apply the Gauss' law to surface S_2 , for which $r \geq R$, we would find that

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}; \text{ (spherical shell, field at } r \geq R \text{)}$$



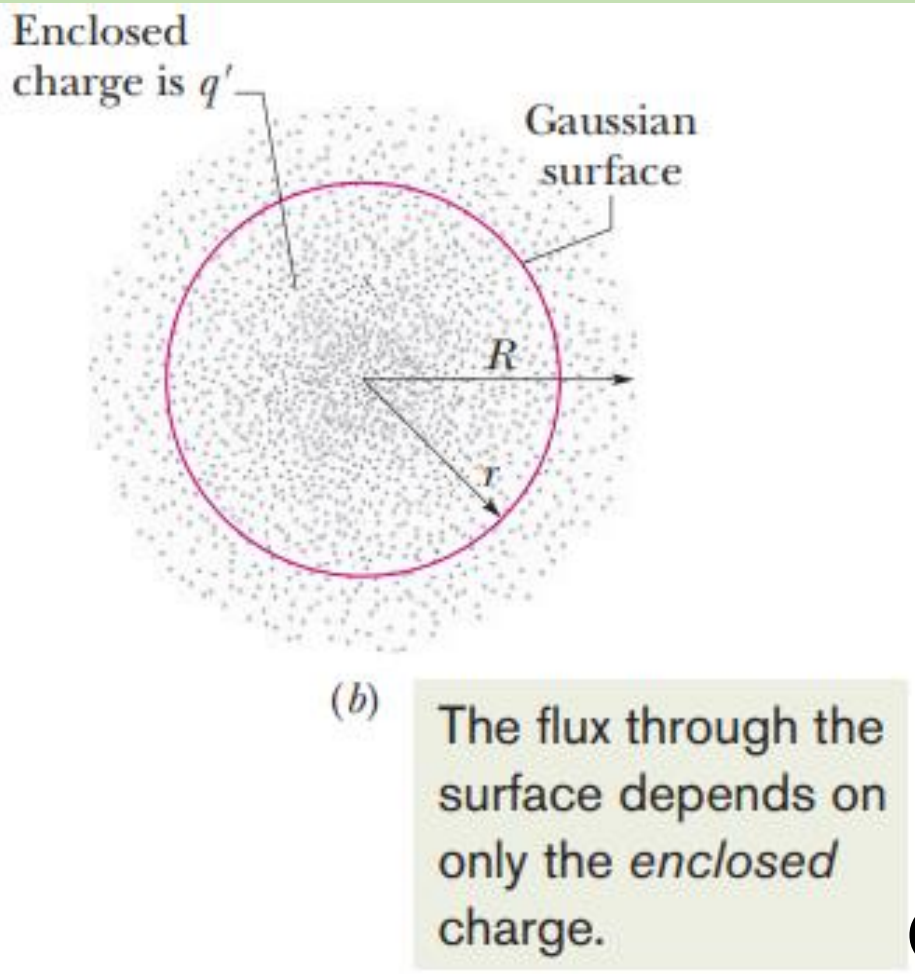
Applying Gauss' law to surface S_1 , for which $r < R$, leads directly to

$$E = 0; \text{ (spherical shell, field at } r < R \text{)}$$

because this Gaussian surface encloses no charge.

Let, the charges enclosed by the Gaussian surface q' (**Fig. b**).

If we apply the Gauss' law, for which $r \leq R$, we would find that



$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow \oint E dA = \frac{q'}{\epsilon_0}$$

$$\Rightarrow E \oint dA = \frac{q'}{\epsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{q'}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2};$$

(spherical shell, field at $r \leq R$)

Let, the volume charge density of the sphere is ρ .

$$\frac{q'}{4\pi r^3} = \frac{q}{4\pi R^3}$$

$$\Rightarrow q' = \frac{qr^3}{R^3}$$

$$\text{So, } E = \left(\frac{q}{4\pi\epsilon_0 R^3} \right) r$$

Chapter 24
Electric Potential

□ Electric Potential Energy

When a charged particle moves from place to place under the electrostatic force of another charged particle (which we assume to remain at rest), the work done by the electrostatic force on the first particle only depends on the starting and finishing points and does not depend on the path taken between the points. We know that a force that has this special property as a **conservative force**. We concluded in the PHY111 course that for a conservative force we could define a **potential energy**.

The difference in the potential energy, ΔU of the system as the object moves from its initial to its final position is equal to the negative of the work done by the force:

$$\Delta U = U_f - U_i = -W_{if} = -\int_i^f \vec{F} \cdot \overrightarrow{ds}$$

We can calculate the change in potential energy when a charge q_2 moves from point **a** to point **b** subject to the force due to another charge q_1 at rest.

We can find the electrostatic force on q_2 due to the charge q_1 when the distance between two charges is r .

The force on q_2 is

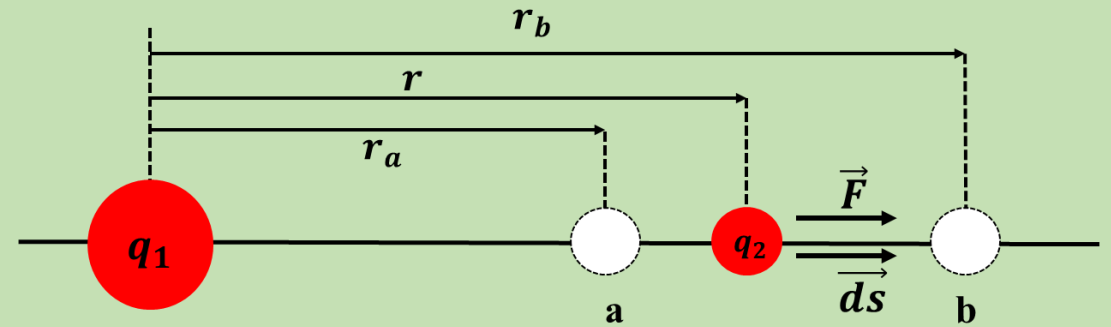
$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

From the definition, the difference in the potential energy, ΔU of the system

$$\Delta U = U_b - U_a = -W_{ab} = -\int_a^b \vec{F} \cdot \vec{ds}$$

$$\Delta U = -\int_a^b \vec{F} \cdot \vec{ds} = -\int_{r_a}^{r_b} F dr = -\frac{q_1 q_2}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{1}{r^2} dr = -\frac{q_1 q_2}{4\pi\epsilon_0} \int_{r_a}^{r_b} r^{-2} dr = -\frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{r^{-2+1}}{-2+1} \right]_{r_a}^{r_b}$$

$$\Delta U = \frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{r_a}^{r_b}; \text{ So, } \Delta U = \frac{q_1 q_2}{4\pi\epsilon_0} \left[\frac{1}{r_b} - \frac{1}{r_a} \right]$$



If $r_a = \infty$, and $r_b = r$, then $U_a = 0$, and $U_b = U$, represents the value of potential energy at point **b**.

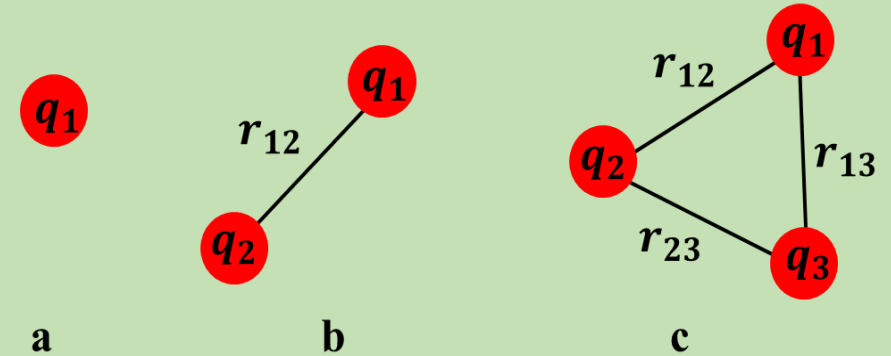
$$U(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

□ Electric Potential Energy of a system of point charges

Let us bring the first charge in from infinity and place it at rest at the location (see **Fig. a**). The potential energy of this configuration is

$$U_1 = 0$$

Since this charge does not interact with any other charges in this process, there is no change in potential energy.



Now, we bring in charge q_2 and fix it at a distance r_{12} from q_1 (See Fig. b). The potential energy of this configuration of q_1 and q_2 is

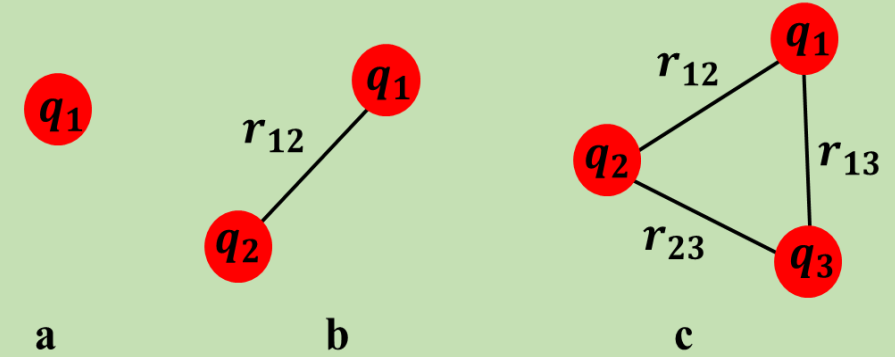
$$U_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Finally, we bring in charge q_3 and place it at a distance r_{13} from q_1 and r_{23} from q_2 (See Fig. c). The potential energy of this configuration is

$$U_3 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

The potential energy of this configuration is

$$\begin{aligned} U &= U_1 + U_2 + U_3 \\ &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}} \end{aligned}$$



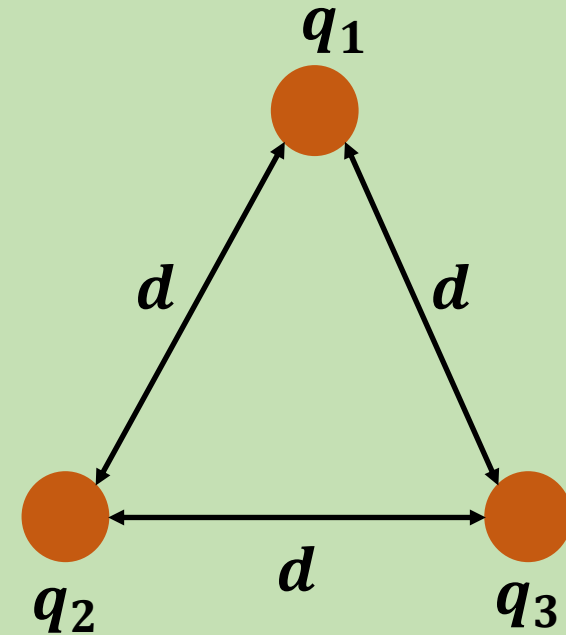
The total potential energy of a system of particles is the sum of the potential energies for every pair of particles in the system.

Q1. The figure shows three charged particles held in fixed positions by forces that are not shown.

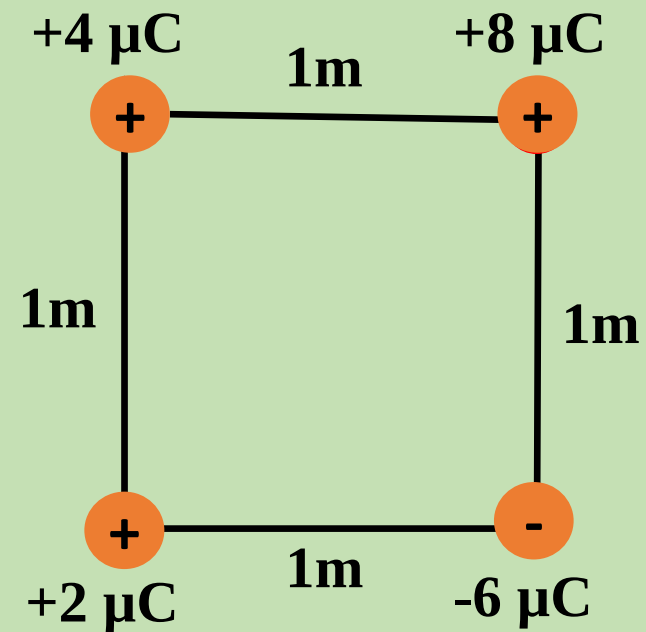
(a) What is the electric potential energy of this system of charges?

Here, $q_1 = +q$, $q_2 = -4q$, and $q_3 = +2q$.

(b) If $d = 12 \text{ cm}$, and $q = 150 \text{ nC}$, then what will be the electric potential energy of this system?



Q2. The figure shows four charged particles held in fixed positions by forces that are not shown. What is the electric potential energy of this system of charges?



Thank You